plusBASIC Quick Reference Guide

Literals

- *Numeric literals* evaluate to floating point or integer numbers.
  - **{+|-}{*digit*...|}{*E*{+|-}*digit*...}** - A floating point decimal number.
  - **{+|-}** - An integer decimal number.
  - **$*hexdigit*...** - An integer hexadecimal number.
  - **'*character*'** - An integer number equal to the ASCII code of *character*.
- *String literals* evaluate to a string of ASCII characters.
  - **""** - A string containing the specified characters.
  - **$""** - A string containing the characters specified by each two-digit hexadecimal number.
  - **\"{{\}*character*}..."** - A string containing C style escape codes.

Variables

- *Numeric variables* contain an floating point or integer numbers.
  - ***alpha*{*alphanum*|~...}** - Numeric variable.
  - ***alpha*{*alphanum*|~...}((*index*{,*index*...})** - Numeric array element.
- *String variables* contain a string of ASCII characters.
  - ***alpha*{*alphanum*|~...}$** - String variable.
  - ***alpha*{*alphanum*|~...}$((*index*{,*index*...})** - String array element.
- *String slices* represent a substring of a string variable.
  - ***var$*[*pos*]** - Character *pos* of string variable *var$*.
  - ***var$*[*start* TO *end*]** - Characters *start* through *end* of string variable *var$*.
  - ***array$(...)*[*pos*]** - Character *pos* of string array element *array$(...)*.
  - ***array$(...)*[*start* TO *end*]** - Characters *start* through *end* of string array element *array$(...)*.

Operators

- *Arithmetic operations* evaluate performed using floating point math.
  - **- *expression*** - Negate expression
  - ***expression* + *expression*** - Add second expression to first expression.
  - ***expression* - *expression*** - Subtract second expression from first expression.
  - ***expression* * *expression*** - Multiply first expression by second expression.
  - ***expression* / *expression*** - Divide first expression by second expression.
  - ***expression* ^ *expression*** - Raise first expression to power second expression.
  - ***expression* MOD *expression*** - Calculate the remainder of the first expression divided by the second expression.
- *Relational operations* return -1 if true, 0 if false.
  - ***expression* > *expression*** - True if first expression is greater than second expression.
  - ***expression* = *expression*** - True if first expression is equal second expression.
  - ***expression* < *expression*** - True if first expression is less than second expression.
  - ***expression* >= *expression*** - True if first expression is greater than or equal to second expression.
  - ***expression* <= *expression*** - True if first expression is less than or equal to second expression.
- *Logical operators* perform the corresponding boolean operation on the bits of the each integer expression.
  - **NOT *integer*** - Sets resulting bit to 1 if the operand bit is 0.
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    - ***integer* AND *integer*** - Sets resulting bit to 1 if both operand bits are 1.
    - ***integer* OR *integer*** - Sets the resulting bit to 1 if either operand bit is 1.
    - ***integer* XOR *integer*** - Sets resulting bit to 1 if one operand bits is 1, and the other is 0.
    - ***integer* IMP *integer*** - *Not implemented*.
- *String operators* create new temporary string.
    - ***string* + *string*** - Concatenates second string to the end of first string.
    - ***string* %% (*expression*{,...})** - Perform string substitution.

Special Characters

- ***_label*** - Line label. Valid at the beginining of a line, or as the operand of a GOTO or GOSUB.
- **'comment** - Shortcut for REM. Valid at the beginning of a line, or after a statement terminating colon.

Direct Mode Aliases

- **/** - Performs the `LIST` command.
- **$** - Performs the `DIR` command.
- **LS** - Performs the `DIR` command.

Statements and Functions

- **ABS(*float*)** - Return absolute value of expression.
- **APPEND *filespec*,@!_ext_addr_@|@{@*page*,}_address_@,*length*** - Append memory to binary file.
- **APPEND *filespec*,^*strvar*** - Append string to file.
- **APPEND *filespec*,*array*{,ASC|ASCX}** - Append string array to file.
- **APPEND ATTRLIST *strvar*=*attr*,...** - Add to string list of sprite and/or tile attributes.
- **APPEND BYTELIST *strvar*=*byte*,...** - Add to a string list of bytes.
- **APPEND INTLIST *strvar*=*integer*,...** - Add to a string list of integers.
- **APPEND PALETTELIST *strvar*=*palette*,...** - Add to a string list of palette numbers.
- **APPEND PSGLIST *strvar*=*register*,*value*{;...}** - *Proposed enhancement*.
- **APPEND RGBLIST *strvar*=*r*,*g*,*b*|*rgbstr*;...** - Add to a string list of RGB values.
- **APPEND TILELIST *strvar*=*tileno*,...** - Add to a string list of tile indexes.
- **ARGS *arg*{,...} {RETURN *var*{,...}}** - Specify subroutine arguments and return variables.
- **ARGS** - Return number of RUN arguments
- **ARGS$(*argno*)** - Return RUN argument number *index*.
- **ASC(*string*{,*pos*})** - Return ASCII value of specified character in *string*.
- **ASC$(*hexstring*)** - Convert hexadecimal string to ASCII string.
- **ATN(*float*)** - Return arctangent of argument.
- **ATTR(*fgcolor*,*bgcolor*)** - Build color attribute byte.
- **ATTR$(*fgcolor*,*bgcolor*,*count*)** - Build color attributes string.
- **BIN$(_long))** - Convert number to into string of binary digits.
- **BIT(*long*,*bit#*)** - Return status of bit in 23 bit number.
- **BIT(*string*,*bit#*)** - Return status of bit in binary string.
- **BYTE(*string*{,*pos*})** - Return signed byte value of specified character in *string*.
- **CALL {@*page*,}*address* {ARGS *hl*,*de*,*bc*}** - Executes machine language routine at address, optionally passing arguments into the associated registers.
- **CALL MODULE** - *Proposed enhancement*.
- **CD** - Display current path or change directory.
- **CD$** - Return current path.
- **CHR$(*byte*)** - Convert ASCII value to string.
- **CIRCLE (*x*,*y*),*radius*{,{,,{,*aspect*}}}** - *Not implemented*.
- **CLEAR {*integer*{,*address*}}** - Clear all variables, optionally allocating string text space and top of BASIC

RAM.

- **CLEAR *_array_{,...}** - Clear numeric or string array data.
- **CLEAR BITMAP {_fgcolor_,_bgcolor_}** - Clear bitmap layer.
- **CLEAR CURSOR** - Temporarily remove cursor from screen.
- **CLEAR KEYS** - Clear keyboard and autotype buffers.
- **CLOAD {{?}_filename_}|*_array_** - Load or verify program or load array data from cassette.
- **CLOSE #_channel_** - Close file.
- **CLS {_fgcolor_,_bgcolor_}** - Clear the text screen.
- **CLS COLOR {_fgcolor_,_bgcolor_}** - Fill Color RAM with current or specified foreground and background colors.
- **COLOR _fgcolor_{,_bgcolor_}** - Set or get bitmap default color(s).
- **COMPARE (_array_,_array_)** - Compare the binary contents of two numeric arrays.
- **COMPARE (_ext_addr_|{@_page_,}_address_,_ext_addr_|{@_page_,}_address_,_length_)** - Compare two blocks of memory.
- **COMPARE (_filespec_,_filespec_)** - *Not implemented*.
- **CONT** - Resume program execution after STOP, Ctrl-C/Escape, or error.
- **COPY** - Copy text screen to serial printer.
- **COPY *_array_ TO *array** - Copy contents of one numeric array to another.
- **COPY *_array_ TO _ext_addr_|{@_page_,}_address_** - Copy numeric array contents to memory.
- **COPY _ext_addr_|{@_page_,}_address_,_length_ TO _ext_addr_|{@_page_,}_address_** - Copy bytes from one section of memory to another.
- **COPY _ext_addr_|{@_page_,}_address_{,_length_(=} TO *_array_** - Copy memory to array contents.
- **COPY _ext_addr_|@_page_,_address_ TO SCREEN {CHR|ATTR}** - Copy paged memory to screen.
- **COPY FILE _filespec_ TO _filespec_** - Copy file.
- **COPY SCREEN {CHR|ATTR} TO _ext_addr_|@_page_,_address_** - Copy screen to paged memory.
- **COPY TILEMAP @page,width,height,x,y** - *Not implemented*.
- **COS(_float_)** - Return cosine of argument.
- **CSAVE _filename_|*_array_** - Save program or array data to cassette.
- **CURSOROFFSET** - Return text screen relative address at cursor location.
- **CURSORX** - Return text screen cursor column.
- **CURSORY** - Return text screen cursor line.
- **DATA _literal_{,...}** - Define data for `READ` statement.
- **DATE$** - Return the current date.
- **DATETIME$** - Return the current date and time.
- **DEC _numvar_** - Decrement numeric variable.
- **DEC(_hexstring_)** - Convert hexadecimal string to long.
- **DEEK(_ext_addr_|@_page_,_address_)** - Read integer from memory.
- **DEF ATTRLIST _strvar_=_attr_,...** - Create a string list of sprite and/or tile attributes.
- **DEF BYTELIST _strvar_=_byte_,...** - Create a string list of bytes.
- **DEF FN _var_(_var_) = _expression_** - Define user-defined function.
- **DEF INTLIST _strvar_=_integer_,...** - Create a string list of integers.
- **DEF PALETTELIST _strvar_=_palette_,...** - Create a string list of palette numbers.
- **DEF PSGLIST _strvar_=_register_,_value_{;...}** - *Proposed enhancement*.
- **DEF RECTLIST _strvar_ = _x1_,_y1_,_x2_,_y2_{;...}** - Create string list of rectangle coordinates
- **DEF RGBLIST _strvar_=_r_,_g_,_b_|_rgbstr_;...** - Create a string list of RGB values.
- **DEF SPRITE _strvar_= _spritle_, _x-offset_, _y-offset_ ;...** - Create a sprite definition.
- **DEF SPRITE _strvar_= (_cols_, _rows_) , _spritle_** - Create a sprite definition.
- **DEF SPRITE _strvar_=^_strvar_** - Create a sprite definition.
- **DEF TILELIST _strvar_=_tileno_,...** - Create a string list of tile indexes.
- **DEF USR=_address_** - Set USR() machine language routine address.
- **DEF USRINT=_address_** - Set user interupt machine language routine address.
- **DEL _filespec_** - Delete file or directory.
- **DIM _array_(_dim_{,...}),...** - Create and allocate one or more arrays.

- **DIM *array*(*dim*{,...})=expr,...** - Create and populate array.
- **DIM(**array*)** - Return number of dimensions in array.
- **DIM(**array*,*dim*)** - Return size of array dimension.
- **DIR** - Display disk directory.
- **DOKE {&#65533;!_ext_addr_&#65533;|&#65533;@*page*,_address_&#65533;,*integer*{;...}** - Write one or more integers to memory.
- **DRAW *command_string*** - *Not implemented.*
- **DRAW TEXT (*col*,*row)*),*string*** - *Not implemented.*
- **DUMP VARS "*filespec*"** - Dump system and BASIC variables to YAML file.
- **EDIT *lineno*** - Edit specified BASIC line.
- **ELSE** - *Not implemented*.
- **END** - Terminate program execution.
- **ENDKEY** - Return key used to exit enhanced INPUT.
- **ERR** - Return last error code.
- **ERRLINE** - Return line number of last error.
- **ERR$** - Return error message of last error.
- **ERR$(*errno*)** - Return error message of specified error code.
- **ERASE **array*{,**array*...}** - Delete arrays.
- **EVAL(*string*)** - Evaluate expression in string.
- **EXP(*float*)** - Return argument to the power of mathematical constant *e*.
- **FILEDATETIME$(*filespec*)** - Return file timestamp.
- **FILEATTR(*filespec*)** - Return file attribute byte.
- **FILEDIR$(*filespec*)** - Extract path from filespec.
- **FILEEXT$(*filespec*)** - Extract extension from filespec.
- **FILELEN(*filespec*)** - Return file length.
- **FILEPOS(*channel*)** - Return current position in open file.
- **FILESTATUS$(*filespec*)** - Return file information as binary string.
- **FILL BYTES {@*page*},*address*,*oount*,*byte*** - Fill block of memory with byte.
- **FILL WORDS {@*page*},*address*,*oount*,*integer*** - Fill block of memory with integer.
- **FILL BITMAP (*x*,*y*){,*color*}** - *Proposed enhancement*.
- **FILL BITMAP {COLOR *fgcolor*, *bgcolor*}** - Fill bitmap and/or colormap with byte and/or color(s).
- **FILL COLORMAP (*x*,y_)-(*x*,*y*) COLOR *fgcolor*, *bgcolor*** - Fill rectangular section of 1bpp colormap.
- **FILL SCREEN {(*x*,y_)-(*x*,*y*)} {CHR _char) {COLOR *fgcolor*,*bgcolor*}** - Fill screen with character and/or colors.
- **FILL TILEMAP {(*x*,y_)-(*x*,*y*)} TILE *tileno*** - Fill tilemap with specified tile, attributes and palette.
- **FLOAT(*string*{,*offset*})** - Convert four consecutive bytes in string to float.
- **FLOAT$(*long*)** - Convert float to four byte binary string.
- **FOR *var*=*expression* TO *expression* {STEP *expression*]** - Initiate a *FOR* loop.
- **FRE(*float*|*string*)** - Return amount of unused variable and array or string space (after forcing garage collection).
- **GET ARGS *var*{,...}** - Read arguments from argument list after GOSUB.
- **GET SCREEN {CHRS|COLORS} (*x*,y_)-(*x*,*y*) **array*|^*strvar*** - Copy text screen clip to array or string variable.
- **GET TILEMAP (*x*,y_)-(*x*,*y*) **array*|^*strvar*** - Copy tilemap clip to array or string variable.
- **GETCHRDEF$(*char*{,*chrset*})** - Return character definition.
- **GETBORDERCHR** - Return screen border character.
- **GETBORDERCHR$** - Return screen border character as string.
- **GETBORDERCOLOR** - Return screen border colors.
- **GETBORDERCOLOR$** - Return screen border colors as string.
- **GETCHRSET** - Return character set currently in use.
- **GETCOLOR** - Return screen printing colors.
- **GETCURSOR** - Return cursor display mode.
- **GETKEY** - Wait for key and return ASCII code.
- **GETKEY$** - Wait for key and return as string.
- **GETKEY(*string*)** - Wait for key and return position in string.

- **GETKEYREPEAT** - Return Direct mode key repeat status.
- **GETLWR** - Wait for key and return lowercased ASCII code.
- **GETLWR$** - Wait for key and return as lowercased string.
- **GETLWR(*string*)** - Wait for key, convert to lowercase, and return position in string.
- **GETPALETTE$(*palette*)** - Return palette contents.
- **GETPALETTE$(*palette*,*index*)** - Return palette entry.
- **GETSOUNDFAST** - Return `SOUND FAST` status.
- **GETSPEED** - Return current turbo setting.
- **GETSPRITE(*spritle*)** - Return spritle properties.
- **GETSPRITE$(*spritedef*)** - Return properties of each spritle in the spritedef.
- **GETSPRITEX(*spritedef*|*spritle*)** - Return sprite or spritle x-position.
- **GETSPRITEY(*spritedef*|*spritle*)** - Return sprite or spritle y-position.
- **GETTILE$(*tileno*)** - Return tile pixels.
- **GETUPR** - Wait for key and return uppercased ASCII code.
- **GETUPR$** - Wait for key and return as uppercased string.
- **GETUPR(*string*)** - Wait for key, convert to uppercase, and return position in string.
- **GOSUB *lineref*{:ARGS *expression*{,...} {RETURN *var*{,...}}** - Branch to subroutine with optional call and return arguments.
- **GOTO *lineref*** - Branch to line specified by line number or label.
- **HEX$(*number*)** - Convert byte or integer into hexadecimal string.
- **HEX$(*string*)** - Convert ASCII string into hexadecimal string.
- **IF *expression* GOTO|THEN *linefref*** - Branch to line or if expression evaluates to true.
- **IF *expression* THEN *statements*** - Execute statements if expression evaluates to true .
- **IN(*port*)** - Read byte from Z80 I/O port.
- **IN$(*port*,*length*)** - Read multiple bytes from Z80 I/O port.
- **INC *numvar*** - Increment numeric variable.
- **INDEX(**array*,*string*)** - Search for string in element of array
- **INMEM({@*page*,}*address*,*string*)** - Search for string in memory.
- **INT(*number*)** - Return expression rounded down to nearest integer.
- **INT(*string*{,*offset*})** - Convert two consecutive bytes in string to signed integer.
- **INKEY** - Return ASCII code of most recently pressed key.
- **INKEY(*string*)** - Return position in string of most recently pressed key.
- **INKEY$** - Return most recently pressed key as string.
- **INPUT {*stringliteral*;}*var*{,...}** - Prompt for keyboard input and assigns entries to variables.
- **INPUT (*col*,*row*),*minlen*,*maxlen*,{INT|STR} *var*** - Enhanced keyboard input routine.
- **INRECT (*rectlist*,*x*,*y*)** - Return index of first rectangle containing coordinates.
- **INSTR({*offset*,}*haystack*,_needle)** - Search for string *needle* in string *haystack*.
- **JOIN *_array_$ INTO *string* DEL *delimiter*** - Join string array elements into delimited string.
- **JOY(*stick*)** - Read one or both game controllers.
- **JOY$(*id*)** - Read wireless game controller.
- **KEY(*keycode*)** - Check if key is currently pressed.
- **KEY(*string*)** - Return position in in string of keycodes of currently pressed key.
- **KEY REPEAT ON|OFF** - Temporarily enable or disable key repeat during program execution.
- **LEFT$(*string*,*len*)** - Return the first up to *len* characters of string.
- **LEN(*string*)** - Return length of string.
- **LEN(*_array_)** - Return total length of array data.
- **{LET} *var*=*expression*** - Define and assigns value to variable.
- **LINE (*x1*,*y1*)-(*x2*,*y2*){,*color*|PRESET|XOR}** - Draw, erase, or toggle line of bloxels or pixels.
- **LINE (*x1*,*y1*)-(*x2*,*y2*),{*color*|PRESET|XOR},RECT** - Draw, erase, or toggle rectangle of bloxels or pixels.
- **LINE (*x1*,*y1*)-(*x2*,*y2*),{*color*|PRESET|XOR},FILL** - Draw, erase, or toggle filled rectangle of bloxels or pixels.
- **LINE INPUT stringvar** - Read an entire line of input from keyboard.
- **LINE INPUT #channel,stringvar** - Read an entire line of input from file.
- **LIST** - Display program lines starting at beginning of program or specified line.

- **LIST *_array_** - Display contents of specified string array.
- **LIST$(_lineno_|NEXT)** - Detokenize program line.
- **LLIST** - Outputs program lines to printer starting at beginning of program or specified line.
- **LLIST *_array_** - Outputs contemts of specified string array to printer.
- **LOAD _filespec_** - Load BASIC program into memory.
- **LOAD _filespec_,  !_ext_addr_  |  {@_page_,}_address_  ** - Load binary file into memory.
- **LOAD _filespec_,*_array_** - Load binary file data into numeric or string array.
- **LOAD _filespec_,*_array_,ASC** - Load multi-line ASCII file into string array.
- **LOAD _filespec_,^_stringvar_** - Load up to 255 characters of file into string variable.
- **LOAD BITMAP _filespec_** - Load bitmap file into Video RAM.
- **LOAD CHRSET _filespec_** - Load character set into alternate character buffer.
- **LOAD CHRSET _filespec_,SWAP** - Load character set into system character buffer.
- **LOAD COLORMAP _filespec_** - Load 1bpp color map file into Video RAM.
- **LOAD DIR {_filespec_,}*_array_** - Load disk directory filenames into string array.
- **LOAD DIR {_filespec_,}}*_array_,ASC** - Load formatted disk directory into string array.
- **LOAD DIR {_filespec_,}*_array_**,BIN - Load binary disk directory into string array.
- **LOAD FNKEYS _filespec_** - Load function key definitions from file.
- **LOAD MODULE** - *Proposed enhancement.*
- **LOAD PALETTE _p_,_filespec_** - Load file data ino palette.
- **LOAD SAMPLE _filespec_,  !_ext_addr_  |  {@_page_,}_address_  ** - Load digitized sound file into paged RAM.
- **LOAD SCREEN _filespec_** - Load text screen from file.
- **LOAD SCREEN ATTR _filespec_** - Load text colors from file.
- **LOAD SCREEN CHR _filespec_** - *Not implemented*.
- **LOAD TILEMAP _filespec_** - Load tile map into Video RAM.
- **LOAD TILESET {INDEX _tileno_|OFFSET _tileofs_} _filespec_** - Load tile definitions to Video RAM.
- **LOAD TRACK _filespec_** - Load tracker file track into tracker player buffer.
- **LOCATE _x_,_y_** - Move text cursor to specified position.
- **LOG(_float_)** - Return natural logarithm of argument.
- **LONG(_string_{,_offset_})** - Convert four consecutive bytes in string to unsigned long.
- **LONG$(_long_)** - Convert long to unsigned four byte binary string.
- **LOOP TRACK** - Play specified or previously loaded tracker file repeatedly.
- **LPOS(_expression_)** - Return current column position of printer.
- **LPRINT TAB(_integer_)|SPC(_integer_)|_expression_|;|,** - Output data to printer.
- **LWR(_char_)** - Lowercases numeric ASCII code or ASCII code of first character of string
- **LWR$(_string_|_byte_)** - Lowercases argument string or argument ASCII code converted to string.
- **LWRKEY** - Return lowercased ASCII code of last key presed.
- **LWRKEY$** - Return lowercased string containing last key pressed.
- **LWRKEY(_string_)** - Convert last key pressed to lowercase, and return position in string.
- **MENU** - *Not implemented.*
- **MID$(_string_,_start_{,_len_})** - Return first up to *len* or all remaining characters starting at character *start* of string.
- **MID$(_strvar_,_start_{,_len_})=_string_** - Replace text in string variable.
- **MKDIR _dirname_** - Create directory.
- **MOUSEB** - Return mouse button state.
- **MOUSEW** - Return mouse wheel delta.
- **MOUSEX** - Return mouse x-position.
- **MOUSEY** - Return mouse y-position.
- **NEXT {_var_,...}** - Terminate one or more *FOR* loops.
- **NEW** - Delete program and clears all variables.
- **OFFSET(_column_,_row_)** - Return offset into Screen or Color RAM.
- **ON _integer_ GOSUB _lineref_,...** - GOSUB one of several line numbers based on *expression*.
- **ON _integer_ GOTO _lineref_,...** - GOTO one of several line numbers based on *expression*.
- **ON ERROR GOTO _lineref_** - Enable or disable error trapping.

- **ON MOUSE GOTO|GOSUB *lineref*** - *Proposed enhancement*
- **ON TIMER GOTO|GOSUB *lineref*** - *Proposed enhancement*
- **OPEN *filespec* FOR INPUT|OUTPUT|APPEND|RANDOM as *numvar*** - Open file.
- **OUT *port*,*byte*|*string*{,...}{;...}** - Write to a Z80 port.
- **OUTPUT #*channel*,*expression*;|,** - Write data to file.
- **PAD$(*string*,*length*{,*char*})** - Pad string on left or right with optional fill character.
- **PAD$(*number*,*length*{,*char*})** - Convert number to padded string.
- **PAUSE** - Halt program execution until key is pressed.
- **PAUSE *jiffies*** - Pauses program execution for the specified amount of time.
- **PAUSE UNTIL *expression*** - Pauses program execution until *expression* evaluates to true.
- **PAUSE TRACK** - Pause currently playing track.
- **PEEK(�!_ext_addr_�|�{@*page*,}_address_�)** - Read byte from memory.
- **PEEK$(�!_ext_addr_�|�{@*page*,}_address_�,*length*)** - Read string from memory.
- **PEEKCOLOR(*address*)** - Read byte from color RAM.
- **PEEKCOLOR$(*address*,*length*)** - Read string from color RAM.
- **PEEKSCREEN(*address*)** - Read byte from screen RAM.
- **PEEKSCREEN$(*address*,*length*)** - Read string from screen RAM.
- **PLAY SAMPLE �!_ext_addr_�|�@*page*,_address_�** - Play digital sample from previously loaded file.
- **PLAY SAMPLE "*filespec*"** - Load and play digitized sound file.
- **PLAY TRACK {*filespec*}** - Play specified or previously loaded tracker file.
- **POINT(*x*,*y*)** - Return status of bloxel or pixel.
- **POKE �!_ext_addr_�|�{@*page*,}_address_�,*byte*|*string*{;..}.** - Write bytes and/or strings to memory.
- **POKE COLOR *address*,*byte*|*string*{;...}** - Write bytes and/or strings to color RAM.
- **POKE SCREEN *address*,*byte*|*string*{;...}** - Write bytes and/or strings to screen RAM.
- **POS(*expression*)** - Return cursor position in current line.
- **POSX** - Return last pixel or bloxel x-position.
- **POSY** - Return 1bpp pixel or bloxel y-position.
- **PRESET (*x*,*y*)** - Erase bloxel or pixel.
- **PRINT @(*x*,*y*)|TAB(*integer*)|SPC(*integer*)|*expression*|;|,** - Display data on screen.
- **PSG *register*,*byte*{,...}** - Write bytes to PSG register
- **PSG *register*,*byte*{;...}** - Not implemented
- **PSET (*x*,*y*){,color}** - Draw bloxel or pixel.
- **PSET (*x*,*y*),XOR** - Toggle bloxel or pixel.
- **PUT SCREEN {CHR|ATTR} (*x*,y_) **array*|^*strvar*** - Copy clip from array or string variable to text screen.
- **PUT TILEMAP (*x*,y_),**array*|^*strvar*** - Copy clip from array or string variable to tilemap.
- **READ *var*{,...}** - Read values from DATA statements into variables.
- **READ **array*{,...}** - Read values from DATA statements into an array.
- **READ #*channel*,^*strvar*,*length*)** - Read string from open file.
- **RECT (*x*,y_)-(*x*,*y*){,{,�_fgcolor_,_bgcolor_�|�_drawcolors_�}}** - Draw rectangle on text screen.
- **REM *text*** - Denotes comment.
- **RENAME *filespec* TO *filespec*** - Rename file or directory
- **RESUME *lineref*** - Clear error condition and continues program execution at specified line
- **RESUME TRACK** - Resume playing paused track.
- **RESET **array*** - Set all array elements to 0 or empty string and collect garbage if string array.
- **RESET BIT *var*,*bit#*** - Set specified bit in numeric ot string variable to 0.
- **RESET BORDER** - Set border character and colors to default values.
- **RESET CHRDEF *char*** - Revert character to default definition.
- **RESET CHRSET *byte*** - *Not implemented*.
- **RESET PALETTE *palette*|*** - Reset single palette or all palettes to default RGB values.
- **RESET SCREEN** - Reset current text screen to default settings.
- **RESET SPRITE *spritedef*|#*spritle*|*** - Reset all spritle properties.
- **RESET USRDEF** - Reset `USR()` vector to default address
- **RESTORE** - Reset READ/DATA pointer to beginning of program or specified line.
- **RESTORE BITMAP|TILEMAP** - *Not implemented.*

- **RESTORE CHRSET** - Copy current character set from system buffer.
- **RESTORE SCREEN** - Copy respective screen buffer to current text screen.
- **RETURN {*expression*{,...}}** - Resume program execution after GOSUB, optionally returning results.
- **RGB(_r_,*g*,*b*_|_*rgbstr*_|_*string*,_delimiter_)** - Return integer representing a single palette entry.
- **RGBB(_r_,*g*,*b*_|_*rgbstr*_|_*string*,_delimiter_)** - Return blue component of a single palette entry.
- **RGBG(_r_,*g*,*b*_|_*rgbstr*_|_*string*,_delimiter_)** - Return green component of a single palette entry.
- **RGBR(_r_,*g*,*b*_|_*rgbstr*_|_*string*,_delimiter_)** - Return red component of a single palette entry.
- **RGB$(_r_,*g*,*b*_|_*rgbstr*_|_*string*,_delimiter_)** - Return string representing a single palette entry.
- **RGBDEC$(*string*{,*char*})** - Convert binary RGB value to decimal red,green, and blue.
- **RGBHEX$(*string*{,*char*})** - Convert binary RGB value to decimal red,green, and blue.
- **RIGHT$(*string*,*len*)** - Return the last up to *byte* characters of string.
- **RND(*integer*)** - Return new or previous psuedo-random number or seeds the random number generator.
- **RUN** - Start program execution at first or specified line.
- **RUN *filespec*** - Load and run specified program, cartridge image, or FPGA core.
- **RUN *filespec arg*** - *Not implemented*.
- **SAVE *filespec*{,ASC|CAQ|BIN}** - Save BASIC program to CAQ or ASCII or raw tokenized file.
- **SAVE *filespec*,_!_ext_addr_|{@*page*,}_address_,*length*** - Save memory to binary file.
- **SAVE *filespec*,*array** {ASC|ASCX} - Save numeric or string array data to file.
- **SAVE *filespec*,^*strvar*** - Save string to file.
- **SAVE BITMAP *filespec*** - Save bitmap screen data from Video RAM to file.
- **SAVE CHRSET *filespec*** - Save curent character set to file.
- **SAVE COLORMAP *filespec*** - Save 1bpp color map from Video RAM to file.
- **SAVE FNKEYS *filespec*** - Save function key definitions to file.
- **SAVE PALETTE *p*,*filespec*** - Save palette contents to file.
- **SAVE SCREEN *filespec*{{,PALETTE},BORDERMAP}** - Save text screen to file.
- **SAVE SCREEN *filespec*,ASC** - Save text screen to text file.
- **SAVE SCREEN ATTR *filespec*** - *Not implemented*.
- **SAVE SCREEN CHR *filespec*** - *Not implemented*.
- **SAVE TILEMAP *filespec*** - Save tile map to file.
- **SAVE TILESET *tileno*,*tilecount*,*filespec*** - Save tile definitions to disk.
- **SCREEN ,,,,** - Set screen mode.
- **SCREEN** - Return screen settings.
- **SCROLL BITMAP** - Not implemented.
- **SCROLL SCREEN** - Not implemented.
- **SET BIT *var*,*bit#*** - Set specified bit in numeric ot string variable to 1.
- **SET BORDER {COLOR *fgcolor*,*bgcolor*}** - Enable color printing mode.
- **SET BREAK ON|OFF** - Enable or disable control-c checking.
- **SET CHRDEF *char*,*string*** - Redefine character.
- **SET COLOR *fgcolor*,*bgcolor*** - Enable color printing mode.
- **SET COLOR OFF** - Disable color printing mode.
- **SET CURSOR ON|OFF** - Enable or disable cursor display.
- **SET FAST ON|OFF** - Enable or disable turbo mode.
- **SET FILE *channel* POS TO *long*** - Set current position in opened file.
- **SET FILE ERROR ON|OFF** - Enable or disable file operations generating BASIC errors.
- **SET FNKEY *key* TO *string*** - Set auto-typed text when function key is pressed.
- **SET KEY *byte*** - Temporarily change keyboard buffer settings.
- **SET KEY REPEAT ON|OFF** - Enable or disable key repeat in direct mode.
- **SET MOUSE ON|OFF** - *Proposed enhancement*.

- **SET MOUSE TILE** - *Proposed enhancement*.
- **SET PALETTE *palette* TO *rgblist*** - Set palette entries.
- **SET SAVE ASC ON|OFF** - Enable or disable forced saving of BASIC programs in ASCII format.
- **SET SCROLL ON|OFF** - Enable or disable scrolling of text screen when PRINTing.
- **SET SOUND FAST ON|OFF_** - Set whether SOUND is affected by turbo mode.
- **SET SPEED *speed*** - Set turbo mode.
- **SET SPRITE #*spritle* {ON|OFF} {POS *x,y*}** - Set spritle properties.
- **SET SPRITE #*spritle* TO *props*** - Set combined spritle properties.
- **SET SPRITE *spritedef* TILECLIP **array*|^*strvar*** - Set sprite to tile indexes and properties in tilemap clip.
- **SET SPRITE *spritedef* TO *proplist*** - Set sprite properties from combined properties list.
- **SET SPRITE * OFF|CLEAR** - Disable or clears all sprites.
- **SET TILE *tileno* TO *tiledata*** - Write tile definition Video RAM.
- **SET TILE_ *tileno* TO CHR *char, fg_color, bg_color*** - Convert character definition to tile data and write to Video RAM.
- **SET TILE *array$** - Set multiple tiles from tile array.
- **SET TILEMAP (*x,y*) TO TILE *tileno*** - Set tilemap cell to tile index, attributes, and palette.
- **SET TILEMAP (*x,y*) TO *integer*** - Set tilemap cell to integer represeting combined tile index, attributes, and palette.
- **SET TILEMAP OFFSET *x,y*** - Scroll tilemap to specified position.
- **SET TRACK FAST {ON|OFF}** - Set tracker playback to 50Hz or 60Hz mode.
- **SET TRACK LOOP {ON|OFF}** - Set loop status of active or paused track.
- **SET TRACK SPEED *hertz*** - Set tracker playback rate.
- **SET USRINT ON|OFF** - Enable disable user interrupt.
- **SGN(*expression*)** - Return signum of expression.
- **SIN(*float*)** - Return sine of argument.
- **SOUND (*duration*,*frequency*)** - Generate a tone through the television speaker.
- **SPLIT *string* INTO **array$* DEL *delimiter*** - Split delimited string into string array elements.
- **SPRITECOL(*spritle*,*spritle*)** - Check for collision between spritles.
- **SPRITECOL(*spritedef*,*spritedef*)** - Check for collision between sprites.
- **SQR(_float)** - Return square root of argument.
- **STASH BITMAP** - *Not implemented*.
- **STASH CHRSET** - Copy current character set to system buffer.
- **STASH SCREEN** - Copy current text screen to respective screen buffer.
- **STASH TILEMAP** - *Not implemented*.
- **STOP** - Interrupt program execution.
- **STOP TRACK** - Stop currently playing track.
- **STR$(*float*)** - Convert floating point number to ASCII string.
- **STRING$(*length*{,*byte*|*string*})** - Create string of repeated character.
- **STRPTR(*strvar*)** - Return address of text for string variable.
- **SWAP BITMAP** - *Not implemented*.
- **SWAP CHRSET** - Swap current character set from character set in system buffer.
- **SWAP SCREEN** - Swap current text screen with respective screen buffer.
- **SWAP TILEMAP** - *Not implemented*.
- **SWAP VARS *var*,*var*** - Swap contents of two variables.
- **TAN(*float*)** - Return tangent of argument.
- **TILEMAP(*x,y*)** - Return tile index and properties in tile cell.
- **TILEMAPX** - Return X-offset of tilemap.
- **TILEMAPY** - Return X-offset of tilemap.
- **TILEOFFSET** - Return first free tile index for current graphics mode.
- **TILEOFFSET(*gfx_mode*)** - Return first free tile index for specified graphics mode.
- **TIME$** - Return the current time.
- **TIMER** - Return timer count..
- **TIMER = *long*** - Set and start jiffy timer..
- **TRACKFAST** - Return -1 if tracker playback is set to fast, else 0.

- **TRACKLOOP** - Return -1 if tracker playback is set to loop, else 0.
- **TRACKSPEED** - Return tracker playback speed in Hertz.
- **TRACKSTATUS** - Return -1 if track is playing, else 0.
- **TRIM$(*string*{,*string*})** - Remove characters from both ends of string.
- **TRIM$(*number*)** - Convert number to string with no leading space.
- **TRIMDIR$(*filespec*)** - Remove path from filespec.
- **TRIMEXT$(*filespec*)** - Remove extension from filespec.
- **TRIML$(*expression*{,*string*})** - Remove characters from left end of string.
- **TRIMR$(*expression*{,*string*})** - Remove characters from right end of string.
- **TRON** - Enable line tracing.
- **TROFF** - Disable line tracing.
- **UPR(*char*)** - Uppercases numeric ASCII code or ASCII code of first character of string
- **UPR$(*string|byte*)** - Uppercases argument string or argument ASCII code converted to string.
- **UPRKEY** - Return uppercased ASCII code of last key presed.
- **UPRKEY$** - Return uppercased string containing last key pressed.
- **UPRKEY(*string*)** - Convert last key pressed to uppercase, and return position in string.
- **USE CHRSET *0|1|filespec*** - Copy specified character set to character RAM.
- **USE SCREEN {*@page,*}*text_mode,gfx_mode*** - *Not implemented.*
- **USR(*float*)** - Call user-defined machine language routine passing argument and returning result.
- **VAL(*string*)** - Convert ASCII string to floating point number.
- **VARDEF(**array{$}*)** - Return True if array is defined.
- **VARDEF(*variable*)** - *Proposed enhancement*.
- **VARPTR(*var{}_|*_array{}*)** - Return address of variable, array element, or array
- **VER(*expression*)** - Return System or plusBASIC version as integer.
- **VER$(*expression*)** - Return System or plusBASIC version as printable string.
- **WAIT *port,xor_mask*{,*and_mask*}** - Pause until specifie bit pattern received from IO port.
- **WORD(*string*{,*offset*})** - Convert two consecutive bytes in string to unsigned integer.
- **WORD$(*integer*)** - Convert integer to unsigned two byte binary string.
- **WRITE *#channel,^strvar*** - Write string to open file.
- **WRITE KEYS *string*** - Write string to auto-type buffer.

PRINT control characters

- **CHR$(8) / $"08"** - Move cursor left, wrapping to previous line, and delete character in the new position.
- **CHR$(10) / $"0A"** - Move cursor down one line or scroll screen up if on last line.
- **CHR$(11) / $"0B"** - Clear screen to default colors and do `SET COLOR OFF` and `SET CURSOR ON`.
- **CHR$(13) / $"0D"** - Move cursor to first column of line.

Error Codes

- **1** - NEXT without FOR
- **2** - Syntax
- **3** - RETURN without GOSUB
- **4** - Out of DATA
- **5** - Illegal quantity
- **6** - Overflow
- **7** - Out of memory
- **8** - Undefined line number
- **9** - Subscript out of range
- **10** - Duplicate definition
- **11** - Division by zero
- **12** - Illegal direct
-

13 - Type mismatch
- **14** - Out of string space
- **15** - String too long
- **16** - String formula too complex
- **17** - Can't continue
- **18** - Undefined user function
- **19** - Missing operand
- **20** - No RESUME
- **21** - RESUME without GOSUB
- **22** - Unprintable
- **23** - Line buffer overflow
- **24** - Statement not implemented
- **25** - Undefined line label
- **26** - ARGS without GOSUB
- **27** - Undimensioned array
- **28** - Too many operands
- **29** - Invalid mode
- **30** - Bad range
- **31** - Empty string
- **32** - String length
- **49** - Bad file
- **50** - File not found
- **51** - Too many files
- **52** - Invalid parameter
- **53** - Input past end
- **54** - File already exists
- **55** - I/O
- **56** - No disk
- **57** - Not empty